

Superiority of the COMB Macro in the COMB-MCM Architecture for Edge Machine Learning

The escalating computational demands of modern artificial intelligence, particularly deep learning algorithms deployed on resource-constrained edge devices, have spurred significant interest in non-traditional computing paradigms. Among these, Compute-in-Memory (CIM) has emerged as a promising approach to circumvent the inherent limitations of the conventional Von Neumann architecture.¹ The fundamental challenge addressed by CIM is the "memory wall," a performance bottleneck arising from the physical separation of processing units and memory, leading to substantial energy expenditure and latency due to constant data transfers.¹ By integrating computational capabilities within or near the memory array, CIM aims to minimize data movement, thereby enhancing both energy efficiency and processing speed.¹ This paradigm shift is particularly relevant for the burgeoning field of edge computing, where power efficiency and real-time processing are critical.¹

General CIM macros represent a diverse landscape of architectural designs and underlying memory technologies.¹ These macros typically aim to perform fundamental operations, most notably multiply-accumulate (MAC), in close proximity to the stored data.¹ This inherent integration offers the significant advantages of enhanced parallelism, enabling simultaneous operations across numerous memory cells¹, and a substantial reduction in energy consumption by minimizing the distance data must travel.¹ CIM macros can be broadly categorized into analog CIM (ACIM) and digital CIM (DCIM).¹ ACIM leverages the physical characteristics of memory cells for analog computations, achieving high efficiency in terms of area and power⁴, while DCIM integrates digital logic within the memory, offering high precision and scalability with semiconductor process advancements.¹ Despite these advantages, general CIM macros face challenges such as susceptibility to noise and process variations in analog implementations, the need for efficient analog-to-digital converters, and limitations in supporting a diverse set of computational operations.⁴

The "COMB-MCM: Computing-on-Memory-Boundary NN Processor with Bipolar Bitwise Sparsity Optimization for Scalable Multi-Chiplet-Module Edge Machine Learning" paper, presented at ISSCC 2022, introduces a novel approach to CIM with its COMB macro.¹¹ The central innovation lies in the Computing-on-Memory Boundary (COMB) architecture, which strategically positions itself as a compromise between the extremes of in-memory and near-memory computation.¹¹ This design philosophy likely aims to harness the benefits of both paradigms while mitigating their inherent drawbacks. Furthermore, the COMB macro is implemented within a Multi-Chiplet

Module (MCM) framework¹¹, a crucial architectural decision for enhancing the scalability and potentially the cost-effectiveness of the processor, especially in the context of increasingly complex edge AI systems and the challenges associated with monolithic chip fabrication in the post-Moore era.¹¹ A distinctive feature of the COMB-MCM processor is its bipolar bitwise sparsity optimization technique¹¹, indicating a sophisticated method to leverage the sparsity inherent in neural networks at a granular bit level, considering both positive and negative values to maximize computational efficiency and minimize power consumption.¹¹ The target application for the COMB-MCM processor is edge machine learning¹¹, where stringent demands for energy efficiency, performance, and cost necessitate innovative architectural solutions. The paper emphasizes that the COMB technique achieves high macro computing energy efficiency and low system power overhead¹¹, suggesting the effectiveness of its "memory boundary" approach in realizing these objectives.

The COMB macro distinguishes itself from general CIM macros through several key innovations and advantages. Notably, it achieves enhanced energy efficiency by employing significant input sparsity optimization.¹¹ For instance, the paper reports an energy efficiency of 46.4 TOPS/W at 50% input sparsity, with even greater efficiency at 90% sparsity.¹¹ While many CIM architectures leverage sparsity to improve performance, the "bipolar bitwise" nature of COMB-MCM's optimization suggests a more refined technique compared to common methods that might focus on coarser levels of sparsity.¹¹ By operating at the bit level and considering both positive and negative values, COMB likely enables a more precise skipping of unnecessary computations, leading to greater energy savings. Research indicates that neural networks inherently possess abundant sparsity, particularly in deeper layers, with feature sparsity in convolutional layers for ImageNet ranging from 44.32% to 89.50%.⁵⁶ The COMB-MCM architecture is likely designed to effectively exploit this inherent characteristic. Furthermore, the COMB approach aims for reduced system power overhead by its very nature as a compromise between in-memory and near-memory computing.¹¹ By strategically positioning processing units closer to memory compared to traditional CIM processors¹¹, COMB seeks to minimize the energy consumed by data movement, a dominant factor in the power consumption of conventional architectures.¹ The utilization of a Multi-Chiplet Module (MCM) design in COMB-MCM also contributes to lower communication costs for local computations.¹¹ By integrating multiple chiplets, each with processing units placed near its memory, the distances for data transfer within a chiplet are reduced, leading to lower energy consumption and latency.¹¹ While inter-chiplet communication can introduce overhead⁵⁵, the COMB architecture likely optimizes the distribution of computational tasks across chiplets to minimize these costs, potentially drawing upon principles of optimal decomposition

and combination.¹¹ Moreover, the COMB-MCM architecture addresses the challenges associated with weight updates in large neural networks.¹¹ The paper acknowledges that the transfer of massive off-chip weight data can impact system performance and that even frequent updates for smaller tasks can consume significant power and introduce latency.¹¹ The "memory boundary" design of COMB likely incorporates mechanisms for more efficient on-chip weight management or strategies to reduce the frequency and volume of off-chip weight data transfers. Some general CIM architectures employ techniques like ping-pong SRAM buffers to handle simultaneous computation and weight updates.⁵⁶ Finally, the bipolar bitwise sparsity optimization is a key differentiator of COMB-MCM.¹¹ This likely involves a more granular approach to exploiting sparsity by considering the individual bits of both positive and negative weight and activation values. By selectively skipping operations involving zero bits, COMB-MCM can potentially achieve higher energy efficiency compared to general CIM macros that might only exploit sparsity at the level of entire weights or activations.

The COMB macro's "Computing-on-Memory Boundary" architecture represents a departure from purely in-memory or near-memory approaches prevalent in general CIM designs.¹¹ This unique hybrid strategy likely aims to strike an optimal balance between the computational efficiency of in-memory processing and the flexibility and control offered by near-memory architectures. The adoption of a Multi-Chiplet Module (MCM) architecture further distinguishes COMB-MCM from many monolithic CIM solutions¹¹, providing enhanced scalability for handling increasingly complex edge AI workloads.¹¹ Furthermore, the dedicated bipolar bitwise sparsity optimization technique sets COMB-MCM apart from general CIM macros that may employ less sophisticated sparsity methods.¹¹ In terms of performance, COMB-MCM demonstrates competitive energy efficiency. While some other CIM macros report higher TOPS/W figures under specific conditions, as illustrated in Table 1, COMB's advantage likely lies in its holistic design that considers not only peak efficiency but also factors like power overhead, scalability, and efficient handling of weight updates, which are crucial for practical edge deployment.

Table 1 provides a comparison of the energy efficiency of COMB-MCM with other CIM macros reported in the research snippets. This comparison highlights that while COMB-MCM achieves a respectable energy efficiency of 46.4 TOPS/W at 50% input sparsity, other CIM architectures, particularly those employing different memory technologies or operating under varying conditions, can achieve higher or lower efficiencies. For instance, digital NVM-based CIM macros show comparable efficiency¹¹, while SRAM-based digital CIM engines can exhibit a wide range depending on

sparsity levels.⁵⁶ Analog CIM approaches, such as the DARAM-based macro, can achieve significantly higher efficiency at lower precision.²⁴ This suggests that COMB-MCM's superiority might not be solely based on maximizing peak TOPS/W but rather on a combination of factors tailored for edge machine learning applications.

The integration of an MCM architecture in COMB-MCM has significant implications for both scalability and cost-effectiveness.¹¹ By utilizing multiple chiplets instead of a single monolithic die, COMB-MCM can overcome the physical limitations and increasing costs associated with fabricating very large chips.¹¹ The smaller size of individual chiplets typically leads to higher manufacturing yields, which directly translates to lower costs.⁵³ Furthermore, the modular nature of MCMs allows for greater flexibility in system design, as different types of chiplets with specialized functionalities can be integrated on the same package. This can accelerate design cycles and potentially enable the reuse of chiplet designs across various products.¹¹ While inter-chiplet communication can introduce challenges such as increased power consumption and latency⁵⁵, the "computing on memory boundary" approach in COMB-MCM likely aims to mitigate these overheads by optimizing the partitioning of computation and memory resources across the chiplets and minimizing the need for frequent high-bandwidth communication between them.¹¹ The emergence of industry standards like UCIe for chiplet interconnections further supports the viability and scalability of the MCM approach.⁵⁹

In conclusion, the COMB macro in the COMB-MCM architecture offers several compelling advantages over general CIM macros for edge machine learning applications. Its innovative "Computing-on-Memory Boundary" approach provides a strategic balance between in-memory and near-memory computation, contributing to high energy efficiency and low power overhead. The integration of an MCM architecture enhances scalability and cost-effectiveness, crucial for deploying complex AI models on edge devices. Moreover, the unique bipolar bitwise sparsity optimization technique allows for fine-grained power savings by effectively exploiting the inherent sparsity in neural networks. While the peak energy efficiency of COMB-MCM might not surpass all other reported CIM macros, its combination of architectural innovations and focus on addressing key challenges in edge computing, such as efficient weight updates and scalable design, positions it as a superior solution for a wide range of edge AI applications. Future research could further explore the specific implementation details of the "memory boundary" concept, optimize the inter-chiplet communication within the MCM framework, and investigate the long-term performance and reliability of the bipolar bitwise sparsity optimization techniques.

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